

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following Java operators can be used with `boolean` variables? (Choose all that apply.)

- A. `==`
- B. `+`
- C. `--`
- D. `!`
- E. `%`
- F. `~`
- G. Cast with `(boolean)`

2. What data type (or types) will allow the following code snippet to compile? (Choose all that apply.)

```
byte apples = 5;
short oranges = 10;
_____ bananas = apples + oranges;
```

- A. `int`
- B. `long`
- C. `boolean`
- D. `double`
- E. `short`
- F. `byte`

3. What change, when applied independently, would allow the following code snippet to compile? (Choose all that apply.)

```
3: long ear = 10;
4: int hearing = 2 * ear;
```

- A. No change; it compiles as is.
- B. Cast `ear` on line 4 to `int`.
- C. Change the data type of `ear` on line 3 to `short`.
- D. Cast `2 * ear` on line 4 to `int`.
- E. Change the data type of `hearing` on line 4 to `short`.
- F. Change the data type of `hearing` on line 4 to `long`.

4. What is the output of the following code snippet?

```
3: boolean canine = true, wolf = true;
4: int teeth = 20;
5: canine = (teeth != 10) ^ (wolf=false);
6: System.out.println(canine+", "+teeth+", "+wolf);
```

- A. true, 20, true
- B. true, 20, false
- C. false, 10, true
- D. false, 20, false
- E. The code will not compile because of line 5.
- F. None of the above.

5. Which of the following operators are ranked in increasing or the same order of precedence? Assume the + operator is binary addition, not the unary form. (Choose all that apply.)

- A. +, *, %, --
- B. ++, (int), *
- C. =, ==, !
- D. (short), =, !, *
- E. *, /, %, +, ==
- F. !, |, &
- G. ^, +, =, +=

6. What is the output of the following program?

```
1: public class CandyCounter {
2:     static long addCandy(double fruit, float vegetables) {
3:         return (int)fruit+vegetables;
4:     }
5:
6:     public static void main(String[] args) {
7:         System.out.print(addCandy(1.4, 2.4f) + ", ");
8:         System.out.print(addCandy(1.9, (float)4) + ", ");
9:         System.out.print(addCandy((long)(int)(short)2, (float)4)); } }
```

- A. 4, 6, 6.0
- B. 3, 5, 6
- C. 3, 6, 6
- D. 4, 5, 6
- E. The code does not compile because of line 9.
- F. None of the above.

7. What is the output of the following code snippet?

```
int ph = 7, vis = 2;
boolean clear = vis > 1 & (vis < 9 || ph < 2);
boolean safe = (vis > 2) && (ph++ > 1);
boolean tasty = 7 <= --ph;
System.out.println(clear + "-" + safe + "-" + tasty);
```

- A. true-true-true
 - B. true-true-false
 - C. true-false-true
 - D. true-false-false
 - E. false-true-true
 - F. false-true-false
 - G. false-false-true
 - H. false-false-false
8. What is the output of the following code snippet?
- ```
4: int pig = (short)4;
5: pig = pig++;
6: long goat = (int)2;
7: goat -= 1.0;
8: System.out.print(pig + " - " + goat);
```
- A. 4 - 1
  - B. 4 - 2
  - C. 5 - 1
  - D. 5 - 2
  - E. The code does not compile due to line 7.
  - F. None of the above.
9. What are the unique outputs of the following code snippet? (Choose all that apply.)
- ```
int a = 2, b = 4, c = 2;
System.out.println(a > 2 ? --c : b++);
System.out.println(b = (a!=c ? a : b++));
System.out.println(a > b ? b < c ? b : 2 : 1);
```
- A. 1
 - B. 2
 - C. 3
 - D. 4
 - E. 5
 - F. 6
 - G. The code does not compile.

10. What are the unique outputs of the following code snippet? (Choose all that apply.)

```
short height = 1, weight = 3;
short zebra = (byte) weight * (byte) height;
double ox = 1 + height * 2 + weight;
long giraffe = 1 + 9 % height + 1;
System.out.println(zebra);
System.out.println(ox);
System.out.println(giraffe);
```

- A. 1
 - B. 2
 - C. 3
 - D. 4
 - E. 5
 - F. 6
 - G. The code does not compile.
11. What is the output of the following code?
- ```
11: int sample1 = (2 * 4) % 3;
12: int sample2 = 3 * 2 % 3;
13: int sample3 = 5 * (1 % 2);
14: System.out.println(sample1 + ", " + sample2 + ", " + sample3);
```
- A. 0, 0, 5
  - B. 1, 2, 10
  - C. 2, 1, 5
  - D. 2, 0, 5
  - E. 3, 1, 10
  - F. 3, 2, 6
  - G. The code does not compile.
12. The \_\_\_\_\_ operator increases a value and returns the original value, while the \_\_\_\_\_ operator decreases a value and returns the new value.
- A. post-increment, post-increment
  - B. pre-decrement, post-decrement
  - C. post-increment, post-decrement
  - D. post-increment, pre-decrement
  - E. pre-increment, pre-decrement
  - F. pre-increment, post-decrement

13. What is the output of the following code snippet?

```
boolean sunny = true, raining = false, sunday = true;
boolean goingToTheStore = sunny & raining ^ sunday;
boolean goingToTheZoo = sunday && !raining;
boolean stayingHome = !(goingToTheStore && goingToTheZoo);
System.out.println(goingToTheStore + "-" + goingToTheZoo
 + "-" +stayingHome);
```

- A. true-false-false
  - B. false-true-false
  - C. true-true-true
  - D. false-true-true
  - E. false-false-false
  - F. true-true-false
  - G. None of the above
14. Which of the following statements are correct? (Choose all that apply.)
- A. The return value of an assignment operation expression can be `void`.
  - B. The inequality operator (`!=`) can be used to compare objects.
  - C. The equality operator (`==`) can be used to compare a `boolean` value with a numeric value.
  - D. During runtime, the `&` and `|` operators may cause only the left side of the expression to be evaluated.
  - E. The return value of an assignment operation expression is the value of the newly assigned variable.
  - F. In Java, `0` and `false` may be used interchangeably.
  - G. The logical complement operator (`!`) cannot be used to flip numeric values.
15. Which operators take three operands or values? (Choose all that apply.)
- A. `=`
  - B. `&&`
  - C. `*=`
  - D. `? :`
  - E. `&`
  - F. `++`
  - G. `/`

16. How many lines of the following code contain compiler errors?

```
int note = 1 * 2 + (long)3;
short melody = (byte)(double)(note *= 2);
double song = melody;
float symphony = (float)((song == 1_000f) ? song * 2L : song);
```

- A. 0
  - B. 1
  - C. 2
  - D. 3
  - E. 4
17. Given the following code snippet, what are the values of the variables after it is executed? (Choose all that apply.)

```
int ticketsTaken = 1;
int ticketsSold = 3;
ticketsSold += 1 + ticketsTaken++;
ticketsTaken *= 2;
ticketsSold += (long)1;
```

- A. ticketsSold is 8.
  - B. ticketsTaken is 2.
  - C. ticketsSold is 6.
  - D. ticketsTaken is 6.
  - E. ticketsSold is 7.
  - F. ticketsTaken is 4.
  - G. The code does not compile.
18. Which of the following can be used to change the order of operation in an expression? (Choose all that apply.)

- A. [ ]
- B. < >
- C. ( )
- D. \ /
- E. { }
- F. " "

19. What is the result of executing the following code snippet? (Choose all that apply.)

```
3: int start = 7;
4: int end = 4;
5: end += ++start;
6: start = (byte)(Byte.MAX_VALUE + 1);
```

- A. start is 0.
  - B. start is -128.
  - C. start is 127.
  - D. end is 8.
  - E. end is 11.
  - F. end is 12.
  - G. The code does not compile.
  - H. The code compiles but throws an exception at runtime.
20. Which of the following statements about unary operators are true? (Choose all that apply.)
- A. Unary operators are always executed before any surrounding numeric binary or ternary operators.
  - B. The `-` operator can be used to flip a `boolean` value.
  - C. The pre-increment operator (`++`) returns the value of the variable before the increment is applied.
  - D. The post-decrement operator (`--`) returns the value of the variable before the decrement is applied.
  - E. The `!` operator cannot be used on numeric values.
  - F. None of the above
21. What is the result of executing the following code snippet?
- ```
int myFavoriteNumber = 8;
int bird = ~myFavoriteNumber;
int plane = -myFavoriteNumber;
var superman = bird == plane ? 5 : 10;
System.out.println(bird + "," + plane + "," + --superman);
```
- A. -7,-8,9
 - B. -7,-8,10
 - C. -8,-8,4
 - D. -8,-8,5
 - E. -9,-8,9
 - F. -9,-8,10
 - G. None of the above